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Valve, fluid control device, pressurizing device, and sphygmomanometer

Abstract

A valve includes a housing, a membrane, and a hole. The housing includes a flat plate having a hole connectable to an external pump, a housing component having a hole connectable to a component to which a fluid is discharged, and a valve chamber held between a housing component including the flat plate and the housing component. The membrane divides a valve chamber into a space facing the flat plate and a space facing the housing component. The hole is connected to the space facing the flat plate and the space facing the housing component. The housing component has a hole that is connectable to an outside of the housing and that allows the fluid to flow out from the valve chamber.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4513766	12/1984	Seaborne	N/A	N/A
4648245	12/1986	Mayer	60/387	F16K 11/022
4856487	12/1988	Furuya	N/A	N/A
2007/0026269	12/2006	Nakakubo	137/859	F16K 99/0001
2013/0178752	12/2012	Kodama	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
S62-141610	12/1986	JP	N/A
H02-7196	12/1989	JP	N/A
3096050	12/2002	JP	N/A
2020-153404	12/2019	JP	N/A
2012/141113	12/2011	WO	N/A
WO-2020084978	12/2019	WO	F04B 39/0027

OTHER PUBLICATIONS

Machine English translation of WO_2020084978_A1 (Year: 2025). cited by examiner
International Search Report for PCT/JP2022/018531 dated Jul. 19, 2022. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This is a continuation of International Application No. PCT/JP2022/018531 filed on Apr. 22, 2022 which claims priority from Japanese Patent Application No. 2021-078258 filed on May 6, 2021. The contents of these applications are incorporated herein by reference in their entireties.

BACKGROUND OF THE DISCLOSURE

Field of the Disclosure

(1) The present disclosure relates to a valve that controls the flow direction of a fluid using a deformable membrane, and a fluid control device that includes the valve.

Description of the Related Art

(2) Patent Document 1 describes a fluid control device that includes a piezoelectric pump and a check valve (valve). The housing of the check valve (valve) has a vent, a communicating hole, a quick-discharge hole, and a check valve hole. The vent is connected to the piezoelectric pump. The communicating hole and the check valve hole are connected to a cuff outside the fluid control device. The quick-discharge hole is connected to the outside of the housing of the check valve.

(3) The housing includes a first wall and a second wall that are parallel to each other. The vent and the check valve hole are formed in the first wall. The communicating hole and the quick-discharge hole are formed in the second wall.

(4) The housing includes a diaphragm. The diaphragm divides the internal space in the housing into a first-wall-side space and a second-wall-side space.

(5) The diaphragm is deformed by the flow of the fluid to perform the switching between gas inflow into the cuff and gas outflow from the quick-discharge hole. Patent Document 1: International Publication No. 2012/141113

BRIEF SUMMARY OF THE DISCLOSURE

(6) However, the device described in Patent Document 1 including a diaphragm (a membrane) that functions as an existing check valve may have unwanted vibrations of the diaphragm (the membrane) when gas is discharged through the quick-discharge hole.

(7) Thus, a possible benefit of the present disclosure is to reduce vibrations of a membrane when gas is discharged through a quick-discharge hole.

(8) The present disclosure provides a valve that includes a housing, a membrane, and a communicating passage. The housing includes a first wall having a first hole connectable to an external pump, a second wall having a second hole connectable to a component to which a fluid is discharged, and a valve chamber held between the first wall and the second wall that oppose each other. The membrane divides the valve chamber into a space nearer the first wall and a space nearer the second wall. The communicating passage connects the space nearer the first wall and the space nearer the second wall to each other.

(9) The second wall includes a third hole that is connectable to the outside of the housing and allows the fluid to flow out from the valve chamber. When viewed in a first direction of the housing in which the first wall, the membrane, and the second wall are arranged, the membrane, an opening surface of the first hole facing the valve chamber, and an opening surface of the third hole facing the valve chamber overlap each other, and the communicating passage overlaps neither the opening surface of the first hole facing the valve chamber nor the opening surface of the third hole facing the valve chamber, and the opening surface of the first hole facing the valve chamber and the opening surface of the third hole facing the valve chamber at least partially overlap each other.

(10) In this structure, at the start when the fluid flows out from the second hole to the third hole, a passage (a first passage) from the second hole to the third hole and a passage (a second passage) from the second hole to the first hole through the communicating passage are formed, the membrane is located in the middle between the first hole and the third hole, and the fluid flows at substantially the same quantity of flow through the first passage and the second passage.

(11) This structure can reduce an immediate change of the passage cross section of the first passage and a change of the velocity of flow. This structure thus reduces attraction of the membrane to the third hole after the membrane is deformed toward the first hole. This operation reduces vibrations of the membrane.

(12) The first passage has a smaller pressure loss than the second passage. Thus, the fluid flows to

the first passage at a higher rate, and the membrane covers the opening of the first hole. Thus, the fluid is discharged through the third hole. The membrane covers the first hole with almost no vibrations, and thus, the fluid is more stably and rapidly discharged through the third hole.
(13) The present disclosure can reduce vibrations of a membrane when gas is discharged through a quick-discharge hole.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS
(1) FIG. **1** is a schematic side cross-sectional view of a structure of a fluid control device according to a first embodiment.
(2) FIG. **2** is a diagram of the fluid control device according to the first embodiment to which a cuff is attached.
(3) FIG. **3**A is a partially enlarged side cross-sectional view of a valve according to the first embodiment, and FIG. **3**B is a perspective plan view illustrating the relationship between each hole and the membrane.
(4) FIG. **4** is a partially enlarged side cross-sectional view of the valve, illustrating the flow of gas to a cuff during a gas supply operation.
(5) FIG. **5**A and FIG. **5**B are partially enlarged side cross-sectional views of the valve, illustrating the flow of gas to a cuff during a gas discharging operation.
(6) FIG. **6** is a schematic side cross-sectional view of a structure of a fluid control device according to a second embodiment.
(7) FIG. **7**A is a partially enlarged side cross-sectional view of a valve according to the second embodiment, and FIG. **7**B is a perspective plan view illustrating the relationship between each hole and the membrane.
(8) FIG. **8**A is a partially enlarged side cross-sectional view of a valve according to a third embodiment, and FIG. **8**B is a perspective plan view illustrating the relationship between each hole and the membrane.
(9) FIG. **9** is a partially enlarged side cross-sectional view of the valve, illustrating the flow of gas from a cuff during a gas discharging operation.
(10) FIG. **10**A is a partially enlarged side cross-sectional view of a valve according to a fourth embodiment, and FIG. **10**B is a partially enlarged side cross-sectional view of the valve, illustrating the flow of gas from a cuff during a gas discharging operation.
(11) FIG. **11**A is a partially enlarged side cross-sectional view of a valve according to a fifth embodiment, FIG. **11**B is a diagram of the flow of gas to a cuff during a gas supply operation, and FIG. **11**C is a partially enlarged side cross-sectional view of the valve, illustrating the flow of gas from a cuff during a gas discharging operation.
(12) FIG. **12** is a schematic side cross-sectional view of a structure of a fluid control device according to a sixth embodiment.
(13) FIG. **13** is a schematic side cross-sectional view of a structure of a fluid control device according to a seventh embodiment.
(14) FIG. **14** is a schematic side cross-sectional view of a structure of a fluid control device according to an eighth embodiment.
(15) FIG. **15** is a schematic side cross-sectional view of a structure of a fluid control device according to a ninth embodiment.
(16) FIG. **16**A and FIG. **16**B are diagrams of derived examples of the positional relationship between a hole connected to a pump, a gas outflow hole, and a membrane.
DETAILED DESCRIPTION OF THE DISCLOSURE
First Embodiment

(17) With reference to the drawings, a valve and a fluid control device according to a first embodiment of the present disclosure are described. FIG. **1** is a schematic side cross-sectional view of a structure of a fluid control device according to a first embodiment. FIG. **2** is a diagram of the fluid control device according to the first embodiment to which a cuff is attached. Throughout the drawings, the shapes of components in each of embodiments including the present embodiment are illustrated in a partially or entirely exaggerated manner for ease of understanding the structures of the valve and the fluid control device. In addition, although gas (for example, air) is used below as an example of a fluid, the structure in the present application is also applicable to the case where a liquid is used as an example of the fluid.
(18) As illustrated in FIGS. **1** and **2**, a fluid control device **10** includes a valve **11** and a pump **12**. The valve **11** and the pump **12** share a common component, and are integrally formed as a unit. A cuff **2** is connected to the valve **11**. The cuff **2** corresponds to "a component to which a fluid is discharged" in the present disclosure.
(19) Schematically, to supply gas to the cuff **2**, the pump **12** is driven to supply gas from the pump **12** into the valve **11**. The gas supplied into the valve **11** is discharged into the cuff **2** through a hole **700** in the valve **11**. To discharge the gas from the cuff **2**, the pump **12** is stopped being driven. The gas from the cuff **2** flows from the hole **700** into the valve **11**, and flows out through a hole **790** to the outside of the valve **11**, in other words, to the outside of the fluid control device **10**.
(20) (Valve **11**)
(21) FIG. **3**A is a partially enlarged side cross-sectional view of the valve according to the first embodiment, and FIG. **3**B is a perspective plan view illustrating the relationship between each hole and the membrane. With reference to FIG. **1**, FIG. **2**, FIG. **3**A, and FIG. **3**B, a specific structure of the valve **11** is described below.
(22) The valve **11** includes a housing component **40**, a housing component **70**, a membrane **80**, and a holding plate **90**.
(23) The housing component **40** includes a flat plate **41** and a side wall **42**, and is formed from a material such as metal or resin. The flat plate **41** corresponds to "a first wall" in the present disclosure.
(24) The side wall **42** is connected to the outer edge of the flat plate **41**. The side wall **42** protrudes from a first main surface of the flat plate **41**. The flat plate **41** and the side wall **42** may be integrally formed or separately formed and joined or bonded together. The flat plate **41** and the side wall **42** define a recess **402**.
(25) The flat plate **41** includes a protrusion **410** protruding from its first main surface to which the side wall **42** is connected. The protrusion **410** has a hole **400** extending through the protrusion **410** in a height direction. The protrusion **410** with this shape having the hole **400** has, for example, a hollow cylindrical shape or a circular shape in a plan view (viewed in a direction perpendicular to the first main surface). The first main surface and a second main surface of the flat plate **41** are connected to each other through the hole **400**. The hole **400** serves as a communicating hole in the valve **11** connected to the pump **12**, and corresponds to "a first hole" in the present disclosure. A flat surface at the boundary where the hole **400** is connected to the valve chamber is an opening surface of the hole **400** facing the valve chamber.
(26) The housing component **70** has a plate shape, and is formed from metal or resin. When viewed in a plan, the housing component **70** has substantially the same shape as the housing component **40**. The housing component **70** corresponds to "a second wall" in the present disclosure.
(27) The housing component **70** has the hole **700**, a hole **710**, and the hole **790**. The hole **700** is open in the first main surface of the housing component **70**. A flat surface at the boundary where the hole **700** is connected to the outside is an outer opening surface of the hole **700**. The hole **710** is open in the second main surface (facing the valve chamber) of the housing component **70**. A flat surface at the boundary where the hole **710** is connected to the valve chamber is an outer opening surface of the hole **710**. The hole **700** and the hole **710** overlap each other in a plan view (when

viewed in a direction perpendicular to the first main surface and the second main surface). The hole **700** and the hole **710** are connected to each other. The hole defined by the hole **700** and the hole **710** connected to each other corresponds to “a second hole” in the present disclosure. The hole **700** corresponds to “a first portion” in the present disclosure, and the hole **710** corresponds to “a second portion” in the present disclosure.
(28) The first end of the hole **790** is open in the second main surface (facing the valve chamber) of the housing component **70**. A flat surface at the boundary where the hole **790** is connected to the valve chamber is an opening surface of the hole **790** facing the valve chamber. The second end of the hole **790** is open in an outer surface of the housing component **70**. A flat surface at the boundary where the hole **790** is connected to the outside is an outer opening surface of the hole **790**. The hole **790** corresponds to “a third hole” in the present disclosure.
(29) The membrane **80** is called a valvular membrane or a diaphragm, and formed from a deformable material, such as a rubber sheet. The membrane **80** has a hole **800** near the outer edge. The hole **800** extends through the membrane **80** in a thickness direction. The hole **800** corresponds to “a communicating passage” in the present disclosure.
(30) The holding plate **90** has a plate shape, and is formed from, for example, metal or resin. The holding plate **90** has an opening portion **900**. The opening portion **900** extends through the holding plate **90** in the thickness direction, and has a shape smaller than the profile of the membrane **80** when viewed in a plan.
(31) The holding plate **90** holds the membrane **80** while allowing the membrane **80** to overlap the opening portion **900**. The holding plate **90** holds the outer edge portion of the membrane **80** while allowing the hole **800** to be connected to the opening portion **900**.
(32) When viewed in a plan, the opening portion **900** in the holding plate **90** is greater than the opening surface of the hole **700** and the opening surface of the hole **790**.
(33) These components are assembled in the following positional relationship to form the valve **11**.
(34) The holding plate **90** and the housing component **70** are sequentially placed on the side of the housing component **40** toward which the side wall **42** protrudes. The holding plate **90** abuts on the side wall **42**. The second main surface of the housing component **70** is connected to the holding plate **90**.
(35) In this structure, the housing component **70** is connected to the housing component **40** while holding the membrane **80** and the holding plate **90** in between and while having its second main surface facing the first main surface of the flat plate **41** of the housing component **40**. Thus, the housing of the valve **11** formed by the housing component **40**, the housing component **70**, and the holding plate **90** defines a valve chamber, or a space to which the recess **402** of the housing component **40** and the opening portion **900** of the holding plate **90** are connected. In other words, the valve chamber is a space held between the flat plate **41** of the housing component **40** and the housing component **70**.
(36) In this structure, the membrane **80** is located in the valve chamber, and between the flat plate **41** of the housing component **40** and the housing component **70**. The membrane **80** divides the valve chamber into a first space facing the housing component **40** (a space defined by the recess **402** or a first-wall-side space) and a second space facing the housing component **70** (a space defined by the opening portion **900** or a second-wall-side space). The first space and the second space are connected to each other with the hole **800** in the membrane **80**.
(37) The hole **400** is connected to the first space defined by the recess **402**, and the hole **710** (the hole **700**) and the hole **790** are connected to the second space defined by the opening portion **900**. More specifically, the hole **400** is located across from the hole **710** (the hole **700**) and the hole **790** with the membrane **80** interposed in between. When the valve **11** is viewed in a plan (viewed in a direction in which the housing component **40**, the membrane **80**, and the housing component **70** are arranged (in the first direction of the valve **11**)), the membrane **80** is located to overlap the hole **400**, the hole **710** (the hole **700**), and the hole **790**.

(38) When the valve **11** is viewed in a plan, the opening surface of the hole **400** facing the valve chamber (the first space) and the opening surface of the hole **790** facing the valve chamber (the second space) overlap each other. More specifically, the opening surface of the hole **790** facing the valve chamber (the second space) completely overlaps the opening surface of the hole **400** facing the valve chamber (the first space). In this case, the complete overlap of the opening surfaces indicates that, for example, as illustrated in FIG. **1**, when the area of the opening surface of the hole **790** is smaller than the area of the opening surface of the hole **400**, the entirety of the opening surface of the hole **790** is located within the range of the opening surface of the hole **400** in a plan view.
(39) When the valve **11** is viewed in a plan, the hole **800** overlaps neither the opening surface of the hole **400** facing the valve chamber, nor the opening surface of the hole **790** facing the valve chamber. Preferably, the hole **800** overlaps the hole **710**.
(40) (Operation of Valve **11**)
(41) As described below, the valve **11** with the above structure performs an operation of supplying gas to the cuff **2** and an operation of discharging the gas from the cuff **2**.
(42) (Operation of Supplying Gas to Cuff **2**)
(43) FIG. **4** is a partially enlarged side cross-sectional view of the valve, illustrating the flow of the gas to the cuff during a gas supply operation.
(44) The pump **12** described below is driven to supply gas to the cuff **2**. Driving the pump **12** causes the gas to flow into the first space of the valve chamber from the pump **12** through the hole **400**. The gas that has flowed in from the hole **400** presses the membrane **80**. Thus, the membrane **80** is deformed to cover the hole **790** in the housing component **70**. The hole **790** is thus closed by the membrane **80**.
(45) The gas that has flowed into the first space flows along the membrane **80** through the first space into the second space through the hole **800**. The gas that has flowed into the second space is discharged out of the valve **11** through the hole **710** and the hole **700**. The hole **700** is connected to the cuff **2**. The gas discharged out of the valve **11** from the hole **700** is supplied to the cuff **2**.
(46) At this time, the hole **790** is closed by the membrane **80**, and the membrane **80** is pressed against the wall surface of the housing component **70** facing the pump chamber with a predetermined pressure from the side facing the first space. Thus, the gas that has flowed into the second space from the hole **800** is discharged from the hole **700** without leaking into the hole **790**. Thus, the gas can be efficiently supplied to the cuff **2**.
(47) (Operation of Discharging Gas from Cuff **2**)
(48) FIG. **5**A and FIG. **5**B are partially enlarged side cross-sectional views of the valve, illustrating a flow of the gas from the cuff during a gas discharging operation. FIG. **5**A illustrates a transient state (an initial state) of the discharge, and FIG. **5**B illustrates a steady state of the discharge.
(49) To discharge the gas from the cuff **2**, driving of the pump **12** described below is stopped. The stop of the pump **12** removes the pressure caused by the gas flowing into the cuff **2**. Thus, the cuff **2** has a higher pressure than the valve chamber. The gas in the cuff **2** thus flows into the second space of the valve chamber through the hole **700** and the hole **710**.
(50) (Transient State)
(51) As illustrated in FIG. **5**A, in the transient state, part of the gas that has flowed into the second space presses the membrane **80** toward the housing component **40**. Thus, the membrane **80** moves away from the wall surface of the housing component **70** facing the pump chamber. The hole **790** is thus connected to the second space.
(52) In the transient state, part of the gas that has flowed into the second space flows into the first space through the hole **800**. In the transient state, the membrane **80** is not in contact with the housing component **40**. Thus, as illustrated in FIG. **5**A, the gas that has flowed into the first space flows through the first space along the membrane **80**, and flows out from the hole **400**.
(53) Thus, in the transient state, the gas flows on both sides of the membrane **80** facing the hole

790 (through the second space) and facing the hole **400**. In a plan view, the hole **790** and the hole **400** overlap each other. In this area, the membrane **80** receives substantially the same pressure from both of the first space and the second space (a pressure balanced state). Particularly, when the quantity of the flow of the gas to the hole **790** and the quantity of the flow of the gas to the hole **400** are the same, the membrane **80** receives the same pressure. Thus, as illustrated in FIG. **5**A, the membrane **80** is in contact with neither the housing component **40** nor the housing component **70**.
(54) More specifically, a passage (a first passage) from the hole **700** to the hole **790** and a passage (a second passage) from the hole **700** to the hole **400** through the hole **800** are formed. This structure can reduce an immediate change of the passage cross section of the first passage and a change of the velocity of flow, and reduce a sharp bend of the membrane **80** toward the first space due to an air flow into the hole **790**. This structure can thus reduce, for example, a bend of the membrane **80** toward the second space in reaction to the membrane **80** colliding against the inner wall surface of the flat plate **41** or the far end of the protrusion **410**, and thus reduce vibrations of the membrane **80**.
(55) When the gas keeps flowing into the second space, this balanced state is lost. More specifically, the hole **710** and the hole **790** are located on the same side of the membrane **80**, and the hole **710** and the hole **400** are located across the membrane **80** from each other. Thus, the pressure loss of the passage from the hole **710** to the hole **790** is smaller than the pressure loss from the hole **710** to the hole **400**. This difference in pressure loss can be adjusted by adjusting, for example, the size or position of the hole **800**.
(56) From the relationship of this pressure loss, the gas more easily flows from the hole **710** to the hole **790**, and flows to the hole **790** at a higher rate as more time passes from the start of the discharge. Thus, the membrane **80** is deformed toward the housing component **40**, and then comes into contact with a far end surface **411** of the protrusion **410** of the housing component **40**. As illustrated in FIG. **5**B, the membrane **80** covers the hole **400**, and thus almost all the gas from the hole **710** flows into the hole **790**. In other words, the membrane **80** is deformed almost linearly toward the inner wall surface of the flat plate **41** or the far end of the protrusion **410** while causing almost no large vibrations.
(57) Thus, the valve **11** can discharge the gas in the cuff **2** to the outside with almost no vibrations of the membrane **80**.
(58) In contrast, in a comparison structure (a structure where the hole **790** and the hole **400** are spaced far apart in a plan view), immediately after being discharged from the cuff **2**, the gas flows into the hole **790**, and due to its velocity of flow, the membrane **80** is attracted to the hole **790** again, and almost entirely closes the hole **790**. Thereafter, the membrane is separated again from the hole **790** by the gas pressure, and collides against the opposing wall surface. The membrane thus vibrates. The comparison structure thus causes unwanted vibration sounds. In addition, repeated covering and uncovering of the hole **790** prevents an increase of the discharge speed.
(59) In contrast, the valve **11** according to the present application can reduce unwanted vibration sounds, and improve the gas discharge speed.
(60) The valve **11** includes the protrusion **410**, but may include no protrusion **410**. Nevertheless, the valve **11** including the protrusion **410** can adjust the distance between the opening surface of the hole **400** and the opening surface of the hole **790** without changing the capacity of the pump chamber. Thus, the valve **11** can control the quantity of the flow and the velocity of the flow of the fluid on both sides of the membrane **80** in the above transient state, and can more reliably reduce the occurrence of vibrations.
(61) In the valve **11**, the hole **710** is widened toward the hole **790** further than the hole **700**. The hole **710** may eliminate the portion widened toward the hole **790**. Nevertheless, the hole **710** including the portion widened toward the hole **790** can reduce the pressure loss of the gas flowing from the hole **700** into the hole **790**. More specifically, the cross section of the opening of the hole **710** that is greater than the cross section of the opening of the hole **700** can reduce the pressure loss

of the gas flowing from the hole **700** to the hole **790**. The cross section of the opening indicates the cross section of the hole **700** and the hole **710** taken along the plane parallel to the opening surfaces of the hole **700** and the hole **710**.

(62) This structure can further enhance the discharge speed of the gas. Preferably, at least part of the portion of the hole **710** widened toward the hole **790** overlaps the hole **400**. Thus, the pressure loss of the gas flowing from the hole **700** to the hole **790** can be further reduced.

(63) In the valve **11**, the hole **400** and the hole **790** overlap each other in a plan view. Thus, the area of the membrane **80** can be further reduced than in a structure where the hole **790** and the hole **400** are spaced apart in a plan view. This structure can achieve the size reduction of the membrane **80**, and the size reduction of the valve **11**.

(64) In the valve **11**, the hole **800** is formed in the membrane **80**. Thus, the hole **800** can be located at a position facing an operation position of the valve **11**, that is, the position where the hole **790** and the hole **400** overlap each other. Thus, the valve **11** can achieve further size reduction.

(65) (Pump **12**)

(66) As illustrated in FIG. **1** and FIG. **2**, the pump **12** includes a main flat plate **21**, a frame **22**, couplers **23**, a piezoelectric element **30**, the flat plate **41** of the housing component **40**, a side wall member **50**, and a lid member **60**. The flat plate **41** of the housing component **40** is shared between the valve **11** and the pump **12**.

(67) The main flat plate **21** is circular in a plan view. The frame **22** is disposed to surround the main flat plate **21**. The multiple couplers **23** have a beam shape, and are disposed between the main flat plate **21** and the frame **22**. The multiple couplers **23** support the main flat plate **21** to allow the main flat plate **21** to vibrate with respect to the frame **22**.

(68) The piezoelectric element **30** is circular in a plan view. The piezoelectric element **30** includes a piezoelectric body and a driving conductor. The piezoelectric element **30** is disposed on the first main surface of the main flat plate **21**. At this time, the center of the piezoelectric element **30** and the center of the main flat plate **21** are aligned with each other. Aligning also includes the case where the centers are misaligned but fall within the range of allowable errors.

(69) The piezoelectric element **30** is distorted with an application of a driving voltage. The main flat plate **21** vibrates with a stress caused by distortion of the piezoelectric element **30**.

(70) The side wall member **50** has an annular shape with a hollow **500**, and is disposed between the frame **22** and the housing component **40**. The side wall member **50** is connected to the frame **22** and the housing component **40**.

(71) A space (the hollow **500**) defined by a structure including the main flat plate **21**, the frame **22**, and the couplers **23**, the housing component **40**, and the side wall member **50** serves as a pump chamber of the pump **12**.

(72) The lid member **60** includes a flat plate portion and a frame portion. The frame portion protrudes from the first main surface of the flat plate portion. The flat plate portion has a hole **600** that extends through the flat plate portion.

(73) The lid member **60** is disposed to cover the main flat plate **21**, and the frame portion of the lid member **60** is connected to the frame **22**.

(74) In this structure, when the main flat plate **21** vibrates in the manner as described above, the gas is sucked into the pump **12** through the hole **600**, and discharged into the valve chamber of the valve **11** through the hole **400**.

(75) In this structure, the fluid control device **10** implements the valve **11** and the pump **12** with a single housing while sharing the flat plate **41** and the hole **400** of the housing component **40**. Thus, the fluid control device **10** can reduce its thickness and its size. In addition, the fluid control device **10** including the valve **11** described above can reduce unwanted vibration sounds, and can more rapidly discharge gas from the cuff **2**.

Second Embodiment

(76) A valve and a fluid control device according to a second embodiment of the present disclosure

are described with reference to the drawings.
(77) FIG. **6** is a schematic side cross-sectional view of a structure of a fluid control device according to the second embodiment. FIG. **7**A is a partially enlarged side cross-sectional view of the valve according to the second embodiment, and FIG. **7**B is a perspective plan view illustrating the relationship between each hole and the membrane.
(78) As illustrated in FIG. **6**, FIG. **7**A, and FIG. **7**B, a fluid control device **10**A according to the second embodiment differs from the fluid control device **10** according to the first embodiment in terms of the structure of a valve **11**A. Other components of the fluid control device **10**A are the same as those of the fluid control device **10**, and thus are not described.
(79) The fluid control device **10**A includes a valve **11**A. The valve **11**A includes a membrane **80**A and a holding plate **90**A.
(80) While being disposed on the housing component **70**, the holding plate **90**A protrudes into an area overlapping the hole **710** in a plan view. In other words, the holding plate **90**A includes a portion that extends inward from the outer edge of the valve chamber. Thus, the holding plate **90**A has an opening portion **900** with a smaller area than that of the holding plate **90**.
(81) The holding plate **90**A has a hole **910** in the protrusion. The hole **910** extends through the holding plate **90**A in the thickness direction. The hole **910** corresponds to “a communicating passage” in the present disclosure.
(82) The membrane **80**A is greater than the opening portion **900** of the holding plate **90**A. The membrane **80**A covers the opening portion **900** without covering the hole **910**. The outer edge portion of the membrane **80**A is fixed to the holding plate **90**A. When the valve **11**A is viewed in a plan, the membrane **80**A overlaps the opening surface of the hole **400** facing the valve chamber, and the opening surface of the hole **790** facing the valve chamber.
(83) The valve **11**A with this structure can reduce vibrations of the membrane **80**A. In addition, the valve **11**A can reduce the area of the membrane **80**A, more specifically, the area of a deformed region of the membrane **80**A. Thus, the membrane **80**A can improve its response speed.
Third Embodiment
(84) A valve and a fluid control device according to a third embodiment of the present disclosure are described with reference to the drawings.
(85) FIG. **8**A is a partially enlarged side cross-sectional view of the valve according to the third embodiment, and FIG. **8**B is a perspective plan view of the relationship between each hole and the membrane.
(86) As illustrated in FIG. **8**A and FIG. **8**B, a fluid control device **10**B according to a third embodiment differs from the fluid control device **10**A according to the second embodiment in terms of the structure of a valve **11**B. Other components of the fluid control device **10**B are the same as those in the fluid control device **10**A, and thus are not described. A membrane **80**B in the drawing is the same as the membrane **80**A, and a holding plate **90**B is the same as the holding plate **90**A, and thus they are not specifically described.
(87) The valve **11**B includes a flat plate **41**B. The flat plate **41**B includes a support member **420**. The support member **420** is formed in the hole **400**. The support member **420** has, for example, a mesh shape in a plan view. In other words, the support member **420** partially closes the hole **400** while allowing the gas to pass through the hole **400**.
(88) The end surface of the support member **420** facing the pump chamber is flush with the far end surface **411** of the protrusion **410**.
(89) FIG. **9** is a partially enlarged side cross-sectional view of the valve, illustrating a flow of the gas from the cuff during a gas discharging operation. FIG. **9** illustrates the steady state of the discharging operation.
(90) As illustrated in FIG. **9**, when the gas flows in from the cuff **2** through the hole **700** and the hole **710**, and the membrane **80**B is deformed to come into contact with the far end surface **411** of the protrusion **410**, the membrane **80**B also comes into contact with the support member **420**, and

is supported by the support member **420**. This structure prevents the membrane **80**B from entering the hole **400**. This structure thus reduces the amount of the deformation of the membrane **80**B, and thus improves the deformation responsiveness of the membrane **80**B to a flow of the gas.

Fourth Embodiment

(91) A valve and a fluid control device according to a fourth embodiment of the present disclosure are described with reference to the drawings.

(92) FIG. **10**A is a partially enlarged side cross-sectional view of the valve according to the fourth embodiment, FIG. **10**B is a partially enlarged side cross-sectional view of the valve, illustrating a flow of the gas from the cuff during a gas discharging operation. FIG. **10**B illustrates the steady state of the discharging operation.

(93) As illustrated in FIG. **10**A and FIG. **10**B, a fluid control device **10**C according to a fourth embodiment differs from the fluid control device **10**B according to the third embodiment in terms of the structure of a valve **11**C. Other components of the fluid control device **10**C are the same as those in the fluid control device **10**B, and thus are not described. A membrane **80**C in the drawing is the same as the membrane **80**B, and a holding plate **90**C is the same as the holding plate **90**B, and thus they are not specifically described.

(94) The valve **11**C includes a flat plate **41**C. The flat plate **41**C includes a support member **420**C. The support member **420**C is disposed in the hole **400**. In a plan view, the support member **420**C has the same shape as the support member **420** of the valve **11**B.

(95) The end surface of the support member **420**C facing the valve chamber is disposed further inward in the hole **400** than the far end surface **411** of the protrusion **410**.

(96) As illustrated in FIG. **10**B, when the gas flows in from the cuff **2** through the hole **700** and the hole **710**, and the membrane **80**C is deformed to come into contact with the far end surface **411** of the protrusion **410**, a portion of the membrane **80**C that overlaps the opening surface of the hole **400** enters the hole **400** to come into contact with the support member **420**C and to be supported by the support member **420**C. At this time, adjusting the position of the end surface of the support member **420**C allows the membrane **80**C to enter the hole **400** by an intended depth. This structure can thus improve the deformation responsiveness of the membrane **80**C to the flow of the gas, and achieve a predetermined discharge speed.

Fifth Embodiment

(97) A valve and a fluid control device according to a fifth embodiment of the present disclosure are described with reference to the drawings.

(98) FIG. **11**A is a partially enlarged side cross-sectional view of the valve according to the fifth embodiment, FIG. **11**B is a diagram of a flow of the gas to the cuff during a gas supply operation, and FIG. **11**C is a partially enlarged side cross-sectional view of the valve, illustrating a flow of the gas from the cuff during a gas discharging operation. FIG. **11**C illustrates the steady state of the discharging operation.

(99) As illustrated in FIG. **11**A, FIG. **11**B, and FIG. **11**C, a fluid control device **10**D according to the fifth embodiment differs from the fluid control device **10**A according to the second embodiment in terms of the structure of a valve **11**D. Other components of the fluid control device **10**D are the same as those in the fluid control device **10**A, and thus are not described. A membrane **80**D in the drawings is the same as the membrane **80**A, and a holding plate **90**D is the same as the holding plate **90**A, and thus is not specifically described.

(100) The valve **11**D includes a flat plate **41**D. The flat plate **41**D includes a communicating hole **430**. When the valve **11**D is viewed in a plan, a first opening of the communicating hole **430** overlaps the opening portion **900**. A second opening of the communicating hole **430** overlaps the hole **910**. In other words, in a plan view, the communicating hole **430** is formed in the flat plate **41**D to extend through to connect the opening portion **900** and the hole **910** in the holding plate **90**D.

(101) The housing component including the flat plate **41**D has a side wall with a small height. In a

static state where no gas flows, the membrane **80**D is in contact with the main surface of the flat plate **41**D facing the pump chamber.
(102) In this structure, to supply gas to the cuff **2**, as illustrated in FIG. **11**B, the membrane **80**D closes the hole **790** to connect the communicating hole **430** to the valve chamber. Thus, the valve **11**D allows the gas to flow into the valve **11**D from the hole **400**, and discharges the gas to the cuff **2** through the communicating hole **430**, the hole **910**, the hole **710**, and the hole **700**.
(103) To discharge the gas from the cuff **2**, as illustrated in FIG. **11**C, the membrane **80**D closes the hole **400** to connect the hole **790** to the valve chamber. Thus, the valve **11**D discharges the gas from the cuff **2** to the outside of the valve **11**D through the hole **700**, the hole **710**, and the hole **790**. At this time, for example, the cross section of the communicating hole **430** is adjusted to implement the state where the pressure on both sides of the membrane **80**D is balanced in the transient state of discharging. Thus, the valve **11**D can reduce vibrations of the membrane **80**D.
Sixth Embodiment
(104) A valve and a fluid control device according to a sixth embodiment of the present disclosure are described with reference to the drawings.
(105) FIG. **12** is a schematic side cross-sectional view of a structure of a fluid control device according to a sixth embodiment.
(106) As illustrated in FIG. **12**, a fluid control device **10**E according to a sixth embodiment differs from the fluid control device **10** according to the first embodiment in terms of the structure of a valve **11**E. Other components of the fluid control device **10**E are the same as those in the fluid control device **10**, and thus are not described.
(107) The fluid control device **10**E includes a valve **11**E. The valve **11**E includes a holding plate **90**E. The holding plate **90**E has, for example, an annular shape. The holding plate **90**E holds the outer edge portion of the membrane **80**. The holding plate **90**E is fixed to the flat plate **41** of the housing component **40**.
(108) When the valve **11**E is viewed in a plan, the holding plate **90**E is fixed while allowing the membrane **80** to overlap the hole **400** and the hole **790**.
(109) In this structure, as in the case of the valve **11**, the valve **11**E can reduce vibrations of the membrane **80**. Specifically, regardless of when the membrane **80** is fixed to the housing component **40** or the housing component **70**, the valve **11**E can reduce vibrations of the membrane **80**.
Seventh Embodiment
(110) A valve and a fluid control device according to a seventh embodiment of the present disclosure are described with reference to the drawings.
(111) FIG. **13** is a schematic side cross-sectional view of a structure of a fluid control device according to a seventh embodiment.
(112) As illustrated in FIG. **13**, a fluid control device **10**F according to a seventh embodiment differs from the fluid control device **10**A according to the second embodiment in terms of the structure of a valve **11**F. Other components of the fluid control device **10**F are the same as those in the fluid control device **10**A, and thus are not described. A membrane **80**F has the same structure as the membrane **80**A, and thus is not specifically described.
(113) The fluid control device **10**F includes the valve **11**F. The valve **11**F includes a holding plate **90**F. The holding plate **90**F has, for example, an annular shape. The holding plate **90**F holds the outer edge portion of the membrane **80**F. The holding plate **90**F is fixed to the flat plate **41** of the housing component **40**.
(114) When the valve **11**F is viewed in a plan, the holding plate **90**F is fixed while allowing the membrane **80**F to overlap the hole **400** and the hole **790**.
(115) The holding plate **90**F has a hole **910**F. The hole **910**F extends through an inner surface and an outer surface of the holding plate **90**F.
(116) As in the case of the valve **11**A, the valve **11**F with this structure can reduce vibrations of the membrane **80**F. More specifically, regardless of when the membrane **80**F is fixed to the housing

component **40** or the housing component **70**, the valve **11**F can reduce the vibrations of the membrane **80**F.

Eighth Embodiment

(117) A valve and a fluid control device according to an eighth embodiment of the present disclosure are described with reference to the drawings.

(118) FIG. **14** is a schematic side cross-sectional view of a structure of the fluid control device according to the eighth embodiment.

(119) As illustrated in FIG. **14**, a fluid control device **10**G according to the eighth embodiment differs from the fluid control device **10**A according to the second embodiment in terms of the structure of a valve **11**G. Other components of the fluid control device **10**G are the same as those of the fluid control device **10**A, and thus are not described.

(120) The fluid control device **10**G includes a valve **11**G. Schematically, the valve **11**G includes multiple functional portions of the valve **11** arranged in parallel to each other.

(121) The valve **11**G includes a housing component **40**G, a housing component **70**G, a membrane **80**G**1**, a membrane **80**G**2**, and a holding plate **90**G.

(122) The housing component **40**G includes a flat plate **41** having a hole **400**G**1** and a hole **400**G**2**. The hole **400**G**1** and the hole **400**G**2** are spaced apart from each other.

(123) The housing component **70**G includes a hole **700**G, a hole **710**G, a hole **790**G**1**, and a hole **790**G**2**. The hole **700**G and the hole **710**G are formed in a center region when the housing component **70**G is viewed in a plan. The hole **790**G**1** and the hole **790**G**2** are formed at positions to hold the hole **700**G and the hole **710**G in between. When the valve **11**G is viewed in a plan, the hole **790**G**1** overlaps the hole **400**G**1**, and the hole **790**G**2** overlaps the hole **400**G**2**.

(124) The holding plate **90**G includes an opening portion **900**G**1**, an opening portion **900**G**2**, and a hole **910**G. The hole **910**G is formed at the center when the holding plate **90**G is viewed in a plan. When the valve **11**G is viewed in a plan, the hole **910**G overlaps the hole **710**G in the housing component **70**G. The opening portion **900**G**1** and the opening portion **900**G**2** are formed at positions to hold the hole **910**G in between. When the valve **11**G is viewed in a plan, the opening portion **900**G**1** overlaps the hole **400**G**1** and the hole **790**G**1**, and partially overlaps the hole **710**G. When the valve **11**G is viewed in a plan, the opening portion **900**G**2** overlaps the hole **400**G**2** and the hole **790**G**2**, and partially overlaps the hole **710**G.

(125) The membrane **80**G**1** is held by the holding plate **90**G to cover the opening portion **900**G**1**. Thus, when the valve **11**G is viewed in a plan, the membrane **80**G**1** overlaps the hole **400**G**1** and the hole **790**G**1**.

(126) The membrane **80**G**2** is held by the holding plate **90**G to cover the opening portion **900**G**2**. Thus, when the valve **11**G is viewed in a plan, the membrane **80**G**2** overlaps the hole **400**G**2** and the hole **790**G**2**.

(127) A hole **800**G is formed between the membrane **80**G**1** and the membrane **80**G**2**. When the valve **11**G is viewed in a plan, the hole **800**G overlaps the hole **910**G. Thus, the hole **910**G and the hole **800**G are connected to each other.

(128) As in the case of the valve **11**, the valve **11**G with this structure can reduce vibrations of the membrane **80**G**1** and the membrane **80**G**2**. The valve **11**G includes a first valve function portion including the hole **400**G**1**, the hole **790**G**1**, and the membrane **80**G**1**, and a second valve function portion including the hole **400**G**2**, the hole **790**G**2**, and the membrane **80**G**2**, between the pump **12** and the hole **700**G and arranged in parallel to each other. Thus, regardless of when either the first valve function portion or the second valve function portion is broken or damaged, the valve **11**G can retain the function of controlling the flow direction.

Ninth Embodiment

(129) A valve and a fluid control device according to a ninth embodiment of the present disclosure are described with reference to the drawings.

(130) FIG. **15** is a schematic side cross-sectional view of a structure of a fluid control device

according to a ninth embodiment.
(131) As illustrated in FIG. **15**, a fluid control device **10**H according to a ninth embodiment differs from the fluid control device **10** according to the first embodiment in that it includes a valve **11**H. The valve **11**H differs from the valve **11** according to the first embodiment in that it eliminates the holding plate **90**.
(132) The valve **11**H includes a housing component **70**H. The outer edge portion of the membrane **80** is fixed to and held by the surface of the housing component **70**H facing the valve chamber.
(133) The valve **11**H of the fluid control device **10**H with this structure can achieve the same effects as the valve **11** of the fluid control device **10**.
(134) (Derived Example of Positional Relationship Between Hole **400**, Hole **790**, and Membrane **80**)
(135) FIG. **16**A and FIG. **16**B illustrate derived examples of the positional relationship between a hole connected to the pump, a gas discharge hole, and a membrane. FIG. **16**A and FIG. **16**B are partially enlarged side cross-sectional views of the valve according to the second embodiment used as an example.
(136) In FIG. **16**A, when the valve **11**A is viewed in a plan, the hole **790** is located closer to the hole **700** than the hole **400**, and the opening surface of the hole **790** facing the valve chamber and the opening surface of the hole **400** facing the valve chamber partially overlap each other.
(137) In FIG. **16**B, when the valve **11**A is viewed in a plan, the hole **400** is located closer to the hole **700** than the hole **790**, and the opening surface of the hole **790** facing the valve chamber and the opening surface of the hole **400** facing the valve chamber do not overlap each other.
(138) As illustrated in FIG. **16**A and FIG. **16**B, when the valve **11**A is viewed in a plan, the hole **400** and the hole **790** overlap a deformable area of the membrane **80**A.
(139) This structure can reduce vibrations of the membrane **80**A. Nevertheless, when the valve **11**A is viewed in a plan, the hole **400** and the hole **790** preferably at least partially overlap each other, and more preferably, completely overlap each other, as in the case of the second embodiment.
(140) The membrane **80**A may have any shape such as a circular shape or a polygonal shape, but preferably overlaps the hole **400** and the hole **790** at the center of its deformable region.
(141) (Example of Device to which Fluid Control Device is Applied)
(142) A fluid control device with the above structure is to be applied to, for example, a pressurizing device. The pressurizing device includes the fluid control device with any of the above structures, and the cuff **2** connected to the hole **700** (refer to FIG. **2**). When the fluid control device causes the gas to flow into the cuff **2**, the gas can apply pressure to, for example, a human body in contact with the cuff **2**.
(143) Such a pressurizing device is usable as a sphygmomanometer. A sphygmomanometer includes the above pressurizing device, and a measuring portion that measures the blood pressure based on the pressure of the cuff.
(144) With the above description, structures of transportation of the gas are described. However, the above structures are applicable to a structure where a fluid other than the gas is used.
(145) The above embodiments may be appropriately combined one with another, and each combination can exert the corresponding effects. **10**, **10**A, **10**B, **10**C, **10**D, **10**E, **10**F, **10**G, **10**H fluid control device **11**, **11**A, **11**B, **11**C, **11**D, **11**E, **11**F, **11**G, **11**H valve **12** pump **21** main flat plate **22** frame **23** coupler **30** piezoelectric element **40**, **40**G housing component **41** flat plate **42** side wall **50** side wall member **60** lid member **70**, **70**G, **70**H housing component **80**, **80**A, **80**B, **80**C, **80**D, **80**F, **80**G**1**, **80**G**2** membrane **90**, **90**A, **90**B, **90**C, **90**D, **90**E, **90**F, **90**G holding plate **400**, **400**G**1**, **400**G**2** hole **420**, **420**C support member **700**, **700**G hole **710**, **710**G hole **790**, **790**G**1**, **790**G**2** hole **800**, **800**G hole **900**, **900**G**1**, **900**G**2** opening surface **910**, **910**G hole **2** cuff

Claims

1. A valve comprising: a housing comprising: a first wall having a first hole connectable to an external pump, a second wall having a second hole connectable to a component to which a fluid is discharged, and a valve chamber held between the first wall and the second wall that oppose each other; at least one membrane dividing the valve chamber into a space facing the first wall and a space facing the second wall; and a communicating passage connecting the space facing the first wall and the space facing the second wall to each other, wherein the second wall has a third hole being connectable to an outside of the housing and allowing the fluid to flow out from the valve chamber, wherein, when viewed in a first direction of the housing in which the first wall, the membrane, and the second wall are arranged: the membrane, an opening surface of the first hole facing the valve chamber, and an opening surface of the third hole facing the valve chamber overlap one another, the communicating passage overlaps neither the opening surface of the first hole facing the valve chamber nor the opening surface of the third hole facing the valve chamber, and the opening surface of the first hole facing the valve chamber and the opening surface of the third hole facing the valve chamber at least partially overlap each other, wherein the first wall comprises a protrusion protruding toward the second wall, and wherein the first hole is provided in the protrusion.
2. The valve according to claim 1, wherein the communicating passage is provided in the membrane.
3. The valve according to claim 1, comprising: a plurality of first holes, a plurality of second holes, and a plurality of membranes.
4. The valve according to claim 1, further comprising: a holding plate extending inward from an outer end of the valve chamber, when viewed in the first direction, and holding the membrane, wherein the communicating passage is provided in the holding plate.
5. The valve according to claim 1, wherein, when viewed in the first direction, the opening surface of the first hole facing the valve chamber and the opening surface of the third hole facing the valve chamber are located while having a smaller one of the first and third holes located within a range of the opening surface of a greater one of the first and third holes.
6. The valve according to claim 5, wherein the second hole comprises a first portion connectable to an outside of the housing and a second portion connectable to the space facing the second wall, and wherein, when viewed in the first direction, the second portion is greater than the first portion, and the first portion overlaps the second portion.
7. The valve according to claim 5, further comprising: a holding plate extending inward from an outer end of the valve chamber, when viewed in the first direction, and holding the membrane, wherein the communicating passage is provided in the holding plate.
8. The valve according to claim 5, wherein the communicating passage is provided in the membrane.
9. The valve according to claim 1, further comprising: a support member provided in the first hole.
10. The valve according to claim 9, wherein an end surface of the support member is flush with the opening surface of the first hole facing the valve chamber.
11. The valve according to claim 9, wherein an end surface of the support member is located further inward in the first hole than the opening surface of the first hole facing the valve chamber.
12. A fluid control device, comprising: the valve according to claim 1; and the pump sharing the first wall, and comprising a pump chamber connected to the first hole.
13. A pressurizing device, comprising: the fluid control device according to claim 12; and a cuff connected to the second hole.
14. A sphygmomanometer comprising: the pressurizing device according to claim 13; and a measuring portion configured to measure a blood pressure based on a pressure in the cuff.
15. A valve comprising: a housing comprising: a first wall having a first hole connectable to an external pump, a second wall having a second hole connectable to a component to which a fluid is

discharged, and a valve chamber held between the first wall and the second wall that oppose each other; at least one membrane dividing the valve chamber into a space facing the first wall and a space facing the second wall; and a communicating passage connecting the space facing the first wall and the space facing the second wall to each other, wherein the second wall has a third hole being connectable to an outside of the housing and allowing the fluid to flow out from the valve chamber, wherein, when viewed in a first direction of the housing in which the first wall, the membrane, and the second wall are arranged: the membrane, an opening surface of the first hole facing the valve chamber, and an opening surface of the third hole facing the valve chamber overlap one another, the communicating passage overlaps neither the opening surface of the first hole facing the valve chamber nor the opening surface of the third hole facing the valve chamber, and the opening surface of the first hole facing the valve chamber and the opening surface of the third hole facing the valve chamber at least partially overlap each other, wherein the second hole comprises a first portion connectable to an outside of the housing and a second portion connectable to the space facing the second wall, and wherein, when viewed in the first direction, the second portion is greater than the first portion, and the first portion overlaps the second portion.
16. The valve according to claim 15, wherein the first wall comprises a protrusion protruding toward the second wall, and wherein the first hole is provided in the protrusion.
17. The valve according to claim 15, further comprising: a holding plate extending inward from an outer end of the valve chamber, when viewed in the first direction, and holding the membrane, wherein the communicating passage is provided in the holding plate.
18. A valve comprising: a housing comprising: a first wall having a first hole connectable to an external pump, a second wall having a second hole connectable to a component to which a fluid is discharged, and a valve chamber held between the first wall and the second wall that oppose each other; at least one membrane dividing the valve chamber into a space facing the first wall and a space facing the second wall; a communicating passage connecting the space facing the first wall and the space facing the second wall to each other; and a holding plate holding the membrane that extends inward from an outer end of the valve chamber, when viewed in a first direction, of the housing in which the first wall, the membrane, and the second wall are arranged, wherein the second wall has a third hole being connectable to an outside of the housing and allowing the fluid to flow out from the valve chamber, wherein, when viewed in the first direction: the membrane, an opening surface of the first hole facing the valve chamber, and an opening surface of the third hole facing the valve chamber overlap one another, the communicating passage overlaps neither the opening surface of the first hole facing the valve chamber nor the opening surface of the third hole facing the valve chamber, the opening surface of the first hole facing the valve chamber and the opening surface of the third hole facing the valve chamber at least partially overlap each other, and wherein the communicating passage is provided in the holding plate.